

SOFTWARE ENGINEERING LAB-3

Component Modelling & Architectural Pattern Selection

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STEP 1 — SCENARIO REVIEW

System: Pharmacy Expiry & Re-order Dispatch Engine

Functional Requirements:

1. FR-001: The system shall enforce FEFO picking order by prioritizing medicine batches with the nearest expiry date during dispensing.
2. FR-002: The system shall record and track the expiry date of each medicine batch stored in the pharmacy inventory.
3. FR-003: The system shall monitor the available stock quantity of each medicine and identify when the quantity falls below its configured safety threshold.
4. FR-004: The system shall automatically generate a purchase order request when a medicine's stock quantity falls below its configured safety threshold.
5. FR-005: The system shall dispatch generated purchase order requests to the appropriate inventory supplier.

Non-Functional Requirements:

1. NFR-001: The system shall automatically generate and dispatch supplier purchase order requests when item stock dips below the configured safety threshold while maintaining performance and security under peak load.
2. NFR-002: The system shall maintain an audit record of medicine stock changes, batch expiry updates, and purchase order activities.

Actors:

- Pharmacy Clerk
- Inventory Supplier

- Pharmacy Manager

Key Challenges:

- Performance: FEFO selection, stock monitoring and purchase-order processing should be timely.
- Security: Inventory and dispensing operations should be performed by authorized users.
- Usability: The Pharmacy Clerk should be able to perform pharmacy operations efficiently.
- Data Consistency: Medicine stock quantities and batch expiry information must remain accurate.
- Auditability: Stock, expiry and purchase-order activities should be recorded for traceability.

Scenario Review Summary:

The Pharmacy Expiry & Re-order Dispatch Engine is an automated stock management system for hospital pharmacies. It tracks medicine batch expiry dates, applies FEFO dispensing, monitors stock levels, generates purchase orders when stock falls below the safety threshold, and dispatches purchase orders to the inventory supplier.

STEP 2 – ARCHITECTURAL STYLE ANALYSIS**1. Layered Architecture****Structure:**

The system is organized into separate layers such as Presentation, Business Logic and Data.

Advantages:

- Provides clear separation of concerns.
- Easy to understand and maintain.
- Suitable for traditional applications.

Disadvantages:

- Can introduce performance overhead between layers.

- May create tight coupling between layers.
- Individual components are difficult to scale independently.

Suitability for Pharmacy System:

High. The pharmacy system has clear responsibilities for user interaction, pharmacy business logic and data management. FEFO dispensing, stock monitoring, re-ordering and audit operations can be organized within these layers.

2. Microservices Architecture

Structure:

The system is divided into small, independently deployable services.

Advantages:

- Allows independent scaling of services.
- Provides fault isolation.
- Supports independent deployment.
- Allows different technologies to be used for different services.

Disadvantages:

- Higher operational complexity.
- Network communication can introduce latency.
- Maintaining data consistency across services can be challenging.

Suitability for Pharmacy System:

Medium. The pharmacy functions could be separated into services such as dispensing, inventory and re-ordering, but this would introduce additional communication and operational complexity for this system.

3. Client-Server Architecture

Structure:

Multiple clients communicate with a centralized server.

Advantages:

- Centralized control.
- Simple deployment.
- Easier to maintain data consistency.

Disadvantages:

- The centralized server can become a single point of failure.
- Can become a scalability bottleneck.
- Provides limited fault tolerance.

Suitability for Pharmacy System:

Medium. Centralized management of pharmacy inventory is useful, but the architecture provides less separation between the different responsibilities of the system.

ARCHITECTURE SELECTION

Selected Architecture: Layered Architecture

We chose Layered Architecture for the Pharmacy Expiry & Re-order Dispatch Engine because the system has clearly separated responsibilities. The presentation layer can handle Pharmacy Clerk interactions, the business layer can handle FEFO dispensing, stock monitoring and purchase-order processing, and the data layer can manage medicine batches, expiry information, stock records and audit records.

Layered Architecture also provides a simple and maintainable structure for the pharmacy system while supporting the existing functional requirements for FEFO dispensing, expiry tracking, stock monitoring, automated purchase-order generation and supplier dispatch.

STEP 3 – ARCHITECTURE SELECTION & COMPONENT IDENTIFICATION

Selected Architecture:

Layered Architecture

Components:

1. Pharmacy Clerk Interface

Responsible for handling interactions with the Pharmacy Clerk, including selecting medicines, viewing dispensing information and confirming dispensing operations.

2. Medicine Inventory & Batch Management

Responsible for maintaining medicine inventory, batch information, expiry dates and available stock quantities.

3. FEFO Dispensing Engine

Responsible for checking available medicine batches, identifying the batch with the nearest valid expiry date and supporting FEFO-based dispensing.

4. Stock Monitoring & Re-order Engine

Responsible for monitoring medicine stock levels, comparing stock against safety thresholds and triggering purchase-order generation when stock falls below the threshold.

5. Purchase Order & Supplier Dispatch

Responsible for generating purchase orders and dispatching them to the appropriate Inventory Supplier.

6. Audit & Records

Responsible for maintaining records of stock changes, batch expiry updates, dispensing activities and purchase-order activities for traceability.

STEP 3B – USE CASE TO COMPONENT MAPPING

UC-01 Manage Medicine Batches

→ Medicine Inventory & Batch Management

UC-02 Track Batch Expiry

→ Medicine Inventory & Batch Management

UC-03 Generate FEFO Dispensing List

→ FEFO Dispensing Engine

UC-04 Dispense Medicine

→ FEFO Dispensing Engine

→ Medicine Inventory & Batch Management

UC-05 Monitor Stock Levels

→ Stock Monitoring & Re-order Engine

UC-06 Generate Purchase Order

→ Stock Monitoring & Re-order Engine

→ Purchase Order & Supplier Dispatch

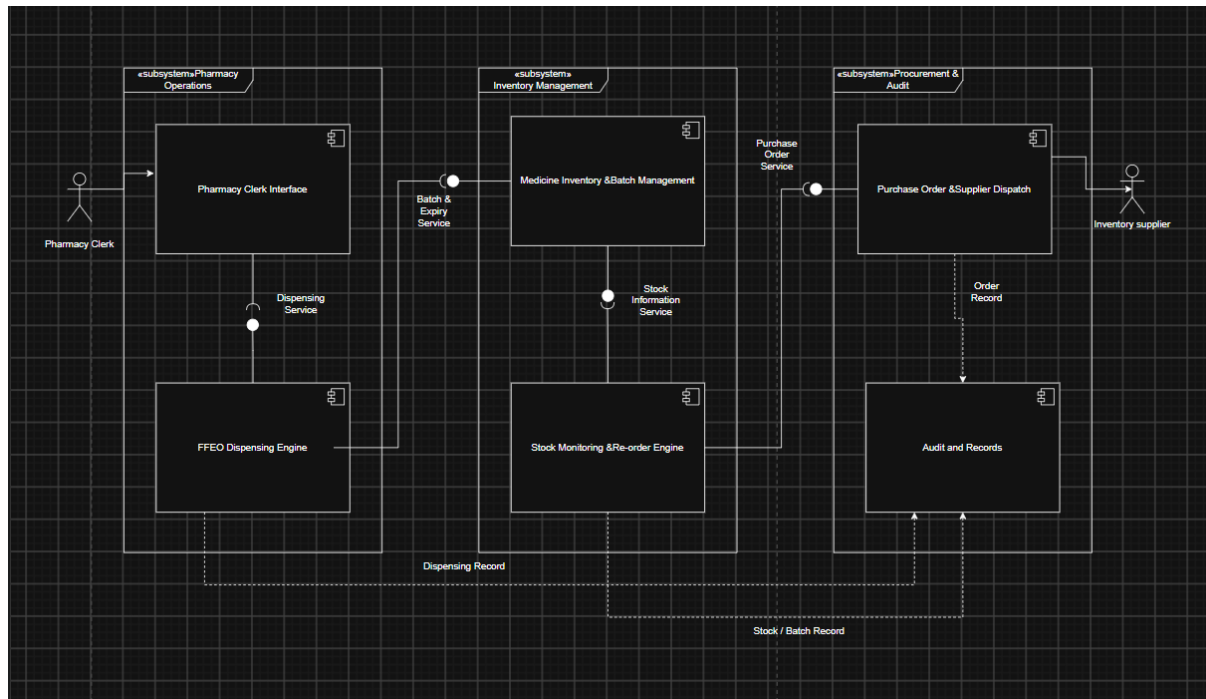
UC-07 Dispatch Purchase Order

→ Purchase Order & Supplier Dispatch

Audit & Records

→ Records activities related to dispensing, stock changes, batch expiry updates and purchase-order activities.

STEP-4 – UML DIAGRAM



STEP 5- Architectural Justification

Selected Architectural Style: Layered Architecture

We chose Layered Architecture for the Pharmacy Expiry & Re-order Dispatch Engine.

The system has clearly separated responsibilities. The Pharmacy Clerk Interface handles user interaction, the FEFO Dispensing Engine and Stock Monitoring & Re-order Engine handle business processing, while Medicine Inventory & Batch Management manages medicine, batch, expiry and stock data.

Layered Architecture is suitable because it provides clear separation of responsibilities, making the system easier to understand and maintain. It supports the major functions of the pharmacy system, including FEFO dispensing, batch expiry tracking, stock-level monitoring, automatic purchase-order generation and supplier dispatch.

The Audit & Records component maintains records of dispensing activities, stock and batch changes, and purchase-order activities, supporting the system's auditability requirement.

